

Quantifying Market Risk: Volatility, VaR and Expected Shortfall

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Introduction

As financial markets are inherently uncertain, managing this uncertainty is the core task in risk management. One of the essential concepts is volatility – a measure of the variability of asset returns, which reflects the level of risk associated with an investment.

The goal of this project is to analyse and model market risk using historical data from a financial asset more specifically, this project focuses on estimating volatility and quantifying potential losses using widely applied risk measures.

Returns were computed based on daily price data and were used to estimate volatility through both historical methods and the Exponentially Weighted Moving Average (EWMA) approach. Then, those volatility estimates were used to calculate Value-at-risk (VaR) and Expected Shortfall (ES), providing insight into potential losses under different market conditions.

The analysis is done in Python and complemented with Excel outputs to facilitate interpretation and visualization. The results highlight the differences between model-based and data-driven approaches to risk measurement, as well as the trade-off between responsiveness and stability in volatility estimation.

Data

The analysis is based on daily price data for the SPDR S&P 500 ETF (SPY), which is used as a proxy for the overall U.S. equity market. The data is obtained using the yfinance Python library and includes adjusted closing prices from 2020 onwards.

To measure market movements, returns are computed from the price series. Two types of returns are considered:

- **Simple returns**, defined as the percentage change in price: $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$
- **Log returns**, defined as: $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$

Although both measures capture price changes, log returns are preferred for volatility modelling since

they are additive over time and are more suitable for statistical analysis.

The return series demonstrate periods of relatively calm behaviour, as well as episodes with high variability, indicating changes in market conditions. These variations form the basis for volatility estimation and risk measurement in the following sections. The presence of volatility clustering, where large movements tend to be followed by large movements, motivates the use of dynamic volatility models.

Volatility models

Volatility is a fundamental measure of financial risk, which represents the variability of returns over time. In this project, two approaches are used to estimate volatility: historical volatility and Exponentially Weighted Moving Average (EWMA) model.

Historical Volatility

Historical volatility is computed as the standard deviation of log returns over a rolling window of 30 days, under the assumption that each observation within the window has equal importance. As a result, it produces a relatively smooth estimate of volatility, however, it may react to sudden changes in market conditions slowly.

EWMA Volatility

To address this limitation, the EWMA model is used. Unlike the historical approach, EWMA assigns more weight to recent observations and less weight to older data. This allows the model to respond more quickly to changes in market volatility.

The EWMA model is defined recursively as:

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

where λ is a smoothing parameter that determines how quickly past information decays. In this analysis, a commonly used value of $\lambda = 0.94$ is applied.

Comparison and Interpretation

There is an important trade-off which was highlighted by comparing the two models. Historical volatility provides a stable estimate, but it may lag during periods of market stress. On the contrary, EWMA is more

responsive to recent shocks and captures sudden changes in risk more effectively.

This difference reflects a broader concept in risk modelling: the balance between stability and responsiveness.

Risk measures

To quantify potential losses, two widely used risk measures are applied: Value-at-Risk (VaR) and Expected Shortfall (ES). These measures provide insight into the downside risk of an investment under both normal and extreme market conditions.

Value-at-Risk (VaR)

Value-at-Risk estimates the maximum expected loss over a given time horizon at a specified confidence level. In this project, a 95% confidence level is used, meaning that losses are expected to exceed the VaR threshold only 5% of the time.

Two approaches to VaR are implemented:

- **Parametric VaR**, which assumes that returns are normally distributed. It is computed using the EWMA volatility estimate:
$$\text{VaR} = -z \cdot \sigma$$

where z is the critical value corresponding to the confidence level.

- **Historical VaR**, which is based on the empirical distribution of past returns. It is calculated as the 5th percentile of the return distribution over a rolling window.

The parametric approach produces smoother estimates that react quickly to changes in volatility, while the historical approach reflects actual past losses but may adjust more slowly.

Expected Shortfall (ES)

While VaR provides a threshold, it does not capture the magnitude of losses beyond that threshold. To address this limitation, Expected Shortfall is used.

Expected Shortfall measures the average loss conditional on losses exceeding the VaR level, providing a more complete view of tail risk.

In this project, ES is computed using historical data by averaging the worst 5% of returns within a rolling window.

Interpretation

The comparison between VaR and Expected Shortfall highlights an important limitation of VaR. Although VaR indicates how often large losses occur, it does not capture how severe those losses can be. Expected Shortfall addresses this by focusing on the tail of the distribution.

This explains why ES has become a more critical component of modern risk management frameworks as a more informative extreme risk measure.

Conclusion

This project shows how market risk can be analysed using historical data and relatively simple quantitative models. By computing returns and estimating volatility through both historical methods and the EWMA approach, it is possible to capture key features of financial time series, such as changing risk levels and volatility clustering.

The results show different risk estimates based on different modelling choices. Historical volatility produces stable estimates and is slow to react to the shocks in the market whereas the EWMA volatility is responsive to shocks due to its assigning greater weights to the most recent prices. The same is true for risk measures such as Value-at-Risk, where parametric and historical models behave differently.

Furthermore, the inclusion of Expected Shortfall highlights the importance of considering not only the frequency of extreme losses but also their magnitude. Therefore, the inclusion of Expected Shortfall takes into account the severity of losses in addition to the likelihood, providing a comprehensive measurement of downside risk and accounting for the paradigm in risk management in finance.

Overall, the project demonstrates a trade-off between simplicity and responsiveness versus realism in risk modelling. The models used in the project are quite simple but still offer a decent level of accuracy and understanding with regards to the market tendency and risk involved.

Possible future enhancements could be more elaborate GARCH models or alternative distributions to better capturing extreme movements in the market.